



COMPUTER ENGINEERING TECHNOLOGY DEPARTMENT

Electrical Circuits Laboratory

Name : _____ Instructor: _____

Section: _____ Date: _____ Score : _____

Title : **Measuring Voltage, Current, and** Lab No. : _____

Resistance

A. Materials and Equipment:

- Carbon Resistors, 1/4 W ~ 1/2W
- Universal PCB
- Soldering Iron, 20W ~ 30W
- Soldering Lead
- Diagonal Cutting Plier
- DC Power Supply
- Multimeter

Resistor Color Code	
0-Black	7-Violet
1-Brown	8-Gray
2-Red	9-White
3-Orange	5%-Gold
4-Yellow	10%-Silver
5-Green	20%-No Color
6-Blue	

B. Resistance Measurement

1. Set the multimeter to the resistance (Ω) mode.
2. Measure the resistance of each resistor individually before installation.
3. Record the measured values in Table I.

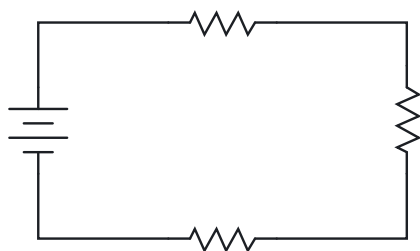
Table I. Resistance Values

Ref	Ideal	Actual
R1		
R2		
R3		

Instruction for Voltage Measurement:

1. Construct the circuit on the universal PCB following the schematic.
2. Set the multimeter to DC voltage (V) mode.
3. Connect the multimeter probes in parallel across the component.
4. Record the measured values in the corresponding data table.

C. Series Circuit





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Table II. Series Circuit Measurements

Ref	Voltage	Current
R1		
R2		
R3		

Using the measured values, calculate the total voltage of the given circuit.

$$v_o =$$

D. Parallel Circuit

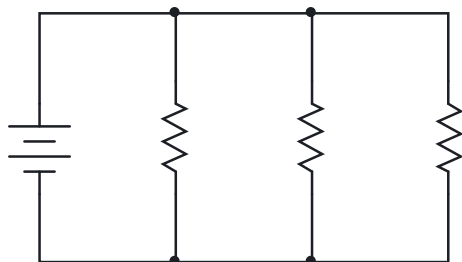


Table III. Parallel Circuit Measurements

Ref	Voltage	Current
R1		
R2		
R3		

Using the measured values, calculate the total current of the given circuit.

$$i_o =$$

E. Series-Parallel Circuit

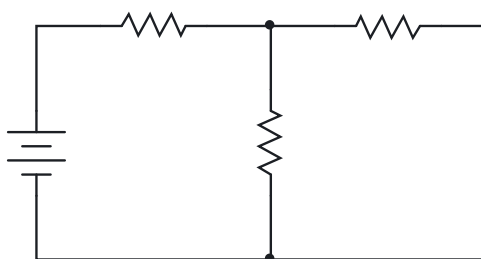


Table IV. Series-Parallel Circuit Measurements

Ref	Voltage	Current
R1		
R2		
R3		

Using the measured values, calculate the total voltage and total current of the given circuit.

$$v_o =$$

$$i_o =$$